University of Puerto Rico at Arecibo
Computer Sciences Department
Computer Sciences Program (B.S.)

SYLLABUS

TITLE	:	Software Engineering
COURSE CODE	:	CCOM 4075
CONTACT HOURS/ CREDITS	:	45 hours. Three (3) hours of conferences per week / Three credits
PREREQUISITES, CO-REQUISITE AND OTHER REQUIREMENTS:	:	MATE 3031, CCOM 3041, CCOM 4006, CCOM 4115

COURSE DESCRIPTION:

This course provides a formal background in software engineering, high quality programming design, specifications, and evaluation. Students are required to participate in a group project. This course will be offered in-person, and hybrid modalities.

LEARNING OBJECTIVES:

Students will gain detailed knowledge on the different phases of software development, such as: analysis, design, coding, testing and maintenance. The student will have the opportunity to work in groups and individually.

At the end of the course, students will have the following competencies:

- Ability to design prototypes to test design concepts
- Competency in determining and producing design specifications of computer software
- Competency in determining the scope of hardware and software development project
- Ability to follow logical and orderly design procedures, choosing the best solution for given set of criteria and considering design constraints and tradeoffs, in the design of software systems
- Ability to articulate teamwork principles
- Be comfortable breaking up a complex problem into separate tasks, delegating tasks to other team members, and integrating the composite effort of the group into a final solution.
- Ability to communicate effectively with other team members

- Ability to effectively describe a problem in a way that can lead to the construction of the solution (Be capable of defining a possible solution)
- Ability to determine the reasonableness of a solution within the physical and ethical context of the problem
- Ability to write effectively and be understood by technical and non- technical audiences
- Be aware of emerging technologies and their impact in the future development of their field of study

TEXTBOOK:

Sommerville. Ian (2015). *Software Engineering*. 10^{th} Edition. Pearson.

COURSE TIME FRAME AND THEMATIC OUTLINE:

Topic	**Time Distribution**	
	In Person	**Hybrid**
I. Introduction to the course	1 week	1 week In person
II. Software Processes	1 week	1 week online
III. Agile Software Development	1 week	1 week online
IV. Requirements Engineering	1 week	1 week online)
V. System Modeling	1 week	1 week online
VI. Architectural Design	1 week	1 week online
VII. Design and Implementation	1 week	1 week online
VIII. Software Testing	1 week	1 week online
IX. Software Evaluation	1 week	1 week online
X. Exams, Lab Sessions and Project Discussions	6 weeks	3 weeks in person 3 weeks online
Total Contact Weeks	**15 weeks**	**15 weeks** (11 online = 73.3% and 4 in person = 26.7%)

INSTRUCTIONAL STRATEGIES:	
It is possible to use some of the following:	
In person	**Hybrid**
• Conferences • Discussion • workshop • Research and presentations • Group Project • Readings of professional articles online	• Online instructional modules • Readings of professional articles online • Instructional videos and audios • Group project • Individual tasks • Asynchronous and synchronous video conferencing

REQUIRED LEARNING RESOURCES:		
Resource	**In person**	**Hybrid**
Account on the institutional learning management platform (e.g. Moodle)	Institution	Institution
Institutional email account	Institution	Institution
Computer with high-speed internet access or mobile device with data service	Student/ Institution	Student
Software: word processor, spreadsheets, presentation editor	Student/ Institution	Student
Integrated or external speakers	Student/ Institution	Student
Webcam or mobile with camera and microphone	Student/ Institution	Student

EVALUATION STRATEGIES:	
In person	**Hybrid**
Exams (3) ....45% Project (3 Phases) ....40% Other ....15% Includes: Oral presentations, homework, quizzes, and other activities **Total 100%**	Exams (3) ....45% Project (3 Phases) .....40% Other ....15% Includes: Oral presentations, homework, quizzes, and other activities **Total 100%**

REASONABLE ACCOMMODATION:
The University of Puerto Rico (UPR) recognizes the right of students with disabilities to an inclusive, equitable and comparable post-secondary education. In accordance with its policy towards students with disabilities, based on federal and state legislation, all qualified students with disabilities have the right to equal participation in those services, programs and activities that are available of a physical, mental or sensory nature, and if this has substantially affected one or more major life activities such as their area of post-secondary studies, they have the right to receive reasonable accommodations or modifications. If you require a reasonable accommodation or modification in this course, you must notify the professor about it, without the need to disclose your condition or diagnosis. Simultaneously, you must request to the Office of Services to Students with Disabilities (OSEI, for its acronym in Spanish) of the unit or Campus, in an expeditious manner, your need for modification or reasonable accommodation.
ACADEMIC INTEGRITY:
The University of Puerto Rico promotes the highest standards of academic and scientific integrity. Article 6.2 of the UPR Students General Bylaws (Board of Trustees Certification 13, 2009-2010) states that academic dishonesty includes, but is not limited to: fraudulent actions; obtaining grades or academic degrees by false or fraudulent simulations; copying the whole or part of the academic work of another person; plagiarizing totally or partially the work of another person; copying all or part of another person answers to the questions of an oral or written exam by taking or getting someone else to take the exam on his/her behalf; as well as enabling and facilitating another person to perform the aforementioned behavior. Any of these behaviors will be subject to disciplinary action in accordance with the disciplinary procedure laid down in the UPR Students General Bylaws. To ensure the integrity and security of user data, any hybrid, distance, and online course should be offered through the institutional learning management platform or by tools required by the course, which uses secure connection and authentication protocols. The system authenticates the user's identity using the username and password assigned in their institutional account. The user is responsible for keeping the password secure, protecting, and not sharing with others.
POLICY AND PROCEDURE FOR HANDLING SITUATIONS OF DISCRIMINATION BASED ON SEX OR GENDER
The Policy and Procedures for Handling Gender Discrimination Situations at the University of Puerto Rico, Certification 107 (2021-2022) Governing Board, ensures that the University of Puerto Rico, as an institution of higher education and workplace, protects the rights and provides a safe environment for all individuals who interact within it, including students, employees, contractors, or visitors. Its purpose is to promote an atmosphere of respect for diversity and the rights of members of the university community. The policy establishes a protocol for handling situations related to the following prohibited behaviors: discrimination based on sex, gender, pregnancy, sexual harassment, sexual violence, domestic violence, dating violence, and stalking, in both the workplace and academic settings.

GRADING SYSTEM:

Quantifiable (letters)

Basic Grading Scale

100% – 90%	**A**
89% – 80%	**B**
79% – 70%	**C**
69% – 60%	**D**
59% and lower	**F**

REFERENCES:

ACM. *Association for Computing Machinery Official Site.* http://www.acm.org/ .

ACM (Association for Computing Machinery). *SIGSOFT* (Special Interest Group on Software engineering, professional journal). http://www.acm.org/sigs/

Pressman, Roger S. (2014). *Software Engineering (A Practitioner's Approach*), 8th Edition. McGraw-Hill Education. 976 p.

Pressman, R.S. & Associates, Inc. *Software Engineering Resources.* www.rspa.com/spi/

Tutorial Freetutes. *Systems Analysis And Design - Complete Introductory Tutorial For Software Engineering*. www.freetutes.com/systemanalysis/

*The instructor will provide additional learning resources throughout the semester.

Reviewed by: Eliana Valenzuela Andrade
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